

BAYESIAN INFERENCES ON THE NONCENTRALITY PARAMETER OF HOTELLING'S - T² STATISTIC

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1. Introduction

Let $X_{\alpha}^{(i)}(px1)$, $\alpha = 1, 2, \dots, N_i$; be independently distributed $N(\mu^{(i)}, \Sigma)$, $i=1, \dots, q$. Let

$$\bar{X}^{(i)} = \sum_{\alpha=1}^{N_i} X_{\alpha}^{(i)} / N_i$$

$$A_i = \sum_{\alpha=1}^{N_i} (X_{\alpha}^{(i)} - \bar{X}^{(i)}) (X_{\alpha}^{(i)} - \bar{X}^{(i)})'$$

$$N = \sum_{i=1}^q N_i$$

$$A = \sum_{i=1}^q A_i$$

$$\bar{X}^* = (\sqrt{N_1} \bar{X}^{(1)}, \sqrt{N_2} \bar{X}^{(2)}, \dots, \sqrt{N_q} \bar{X}^{(q)})'$$

$$M = (\sqrt{N_1} \mu^{(1)}, \sqrt{N_2} \mu^{(2)}, \dots, \sqrt{N_q} \mu^{(q)})'$$

$$F = \bar{X}^{*'} A^{-1} \bar{X}^*$$

$$\Omega = M' \Sigma^{-1} M.$$

The density function of F when $q \leq p \leq N$ is given by James (1964) as

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$$(1.1) f_1(F|\Omega) = (\Gamma_q(\frac{1}{2}N)/\Gamma_q(\frac{1}{2}(N-p)))\Gamma_q(\frac{1}{2}p) \text{etr}(-\frac{1}{2}\Omega) \\ \cdot |F|^{\frac{1}{2}(p-q-1)}|I+F|^{-\frac{1}{2}N} {}_1F_1(\frac{1}{2}N; \frac{1}{2}p; \frac{1}{2}\Omega F(I+F)^{-1}), \quad F > 0$$

Using the improper joint density on M and Σ^{-1} the posterior density function of Ω is given by

$$(1.2) \xi_1(\Omega|F) = \Gamma_q^{-1}(\frac{1}{2}p)2^{-\frac{1}{2}pq} \text{etr}(-\frac{1}{2}\Omega)|\Omega|^{\frac{1}{2}(p-q-1)} \\ \cdot |I+F|^{-\frac{1}{2}(N-q)} {}_1F_1(\frac{1}{2}(N-q); \frac{1}{2}p; \frac{1}{2}\Omega F(I+F)^{-1}), \quad \Omega > 0$$

This density can be proved in the same way as the one given by de Waal (1974) for the case $p \leq q \leq N$ on the roots of Ω by interchanging p and q and using the fact that Ω given Σ^{-1} and $\bar{X}^*$ is distributed $W(I_q, p, \bar{X}^*, \Sigma^{-1}\bar{X}^*)$.

In the same way it follows from the above reference that for a quadratic loss function the Bayes estimate of the j -th elementary symmetric function (esf) of Ω , $tr_j \Omega$, is given by

$$(1.3) E(tr_j \Omega|F) = \sum_{k=0}^j (N-q)^{(k)} (p-k)^{(j-k)} \binom{q-k}{j-k} tr_k F$$

It is easy to show that

$$(1.4) E(\Omega|F) = p I_q + (N-q)F$$

if we use the fact that M conditional on Σ^{-1} and $\bar{X}^*$ is distributed $N(\bar{X}^*, \Sigma \otimes I_q)$. This is also a Bayes estimate for Ω if a quadratic loss is considered.

The following can also be proved:

Theorem 1

$$(1.5) E((\Omega - E\Omega)^2|F) \cong p(q+1)I_q + (N-q)(q+2)F + (N-q)(3+p+N-q)F^2/p \\ + (N-q)(N-q+p+1)F \text{tr} F/p + (N-q)I_q \text{tr} F.$$

Proof.

It is known (Anderson (1958)) that if $A(p \times p)$ is distributed $W(\Sigma, n)$, then

$$E(a_{ij} a_{kl}) = n^2 \sigma_{ij} \sigma_{kl} + n \sigma_{ik} \sigma_{jl} + n \sigma_{il} \sigma_{jk}.$$

$$\text{Hence } E a_{ij}(a_{11} + a_{22} + \dots + a_{pp}) = n^2 \sigma_{ij} \sum_{k=1}^p \sigma_{kk} + 2n \sum_{k=1}^p \sigma_{ik} \sigma_{kj}$$

and therefore

$$(1.6) \quad E A \operatorname{tr} A = n^2 \Sigma \operatorname{tr} \Sigma + 2n \Sigma^2$$

It is also shown by Nel (1972) that

$$(1.7) \quad E(A^2) = n(n+1)\Sigma^2 + n\Sigma \operatorname{tr} \Sigma.$$

If A is however distributed as a noncentral Wishart $W(I, n, T)$ where T is the noncentrality parameter, then it is shown (Steyn and Roux (1972)) that A is approximately distributed $W(\Sigma^*, n)$ where $\Sigma^* = I + \frac{1}{n}T$.

We have that conditional on Σ^{-1} and $\bar{X}^*$ that

$$\Omega | \Sigma^{-1}, \bar{X}^* \text{ is distributed } W(I_q, p, \bar{X}^{*'} \Sigma^{-1} \bar{X}^*)$$

Using the approximation by Steyn and Roux (1972) and (1.7) it follows that

$$(1.8) \quad E(\Omega^2 | \Sigma^{-1}, \bar{X}^*) \cong p(p+1)\Phi^2 + p\Phi \operatorname{tr} \Phi$$

where

$$\Phi = I_q + \frac{1}{p} \bar{X}^{*'} \Sigma^{-1} \bar{X}^*.$$

Since $\Sigma^{-1} | A$ is distributed $W(A^{-1}, N-q)$, (see de Waal (1974)) a posteriori

$$\bar{X}^{*'} \Sigma^{-1} \bar{X}^* \text{ is distributed } W(F, N-q).$$

Hence

$$(1.9) \quad E \Phi^2 = I_q + \frac{2}{p} (N-q)F + \frac{1}{p^2} (N-q+1)F^2 + \frac{N-q}{p^2} F \operatorname{tr} F$$

and

$$(1.10) \quad E \Phi \operatorname{tr} \Phi = qI_q + \frac{q}{p} (N-q)F + \frac{1}{p^2} (N-q)^2 F \operatorname{tr} F + \frac{2(N-q)}{p^2} F^2 + \frac{N-q}{p} I_q \operatorname{tr} F$$

It now follows from (1.8) that

$$(1.11) \quad E(\Omega^2 | F) \cong p(p+q+1)I_q + (N-q)(2p+q+2)F + \frac{N-q}{p} (N-q+p+1)F \operatorname{tr} F \\ + \frac{N-q}{p} (2 + (N-q+1)(p+1))F^2 + (N-q) I_q \operatorname{tr} F$$

and hence the theorem follows.

If we let $q = 1$, F becomes

$$T^2 = N \bar{X}' A^{-1} \bar{X}$$

i.e. $(N-1)^{-1}$ times Hotelling's $-T^2 (= N \bar{X}' S^{-1} \bar{X})$ statistic. T^2 can therefore also be considered as a Hotelling's $-T^2$ statistic as well as F for the multidimensional case. The noncentrality parameter becomes for $q = 1$

$$\tau^2 = N \mu' \Sigma^{-1} \mu$$

with posterior density function

$$(1.12) \quad \xi_2(\tau^2 | T^2) = (1/\Gamma(\frac{1}{2}p)) 2^{-\frac{1}{2}p} e^{-\frac{1}{2}\tau^2} (\tau^2)^{\frac{1}{2}p-1} \\ (1+T^2)^{-\frac{1}{2}(N-1)} {}_1F_1(\frac{1}{2}(N-1); \frac{1}{2}p; \frac{1}{2}\tau^2 T^2 (1+T^2)^{-1}), \tau^2 > 0$$

The density function of T^2 given τ^2 is given by

$$(1.13) \quad f_2(T^2 | \tau^2) = (\Gamma(\frac{1}{2}N) \Gamma(\frac{1}{2}(N-p)) \Gamma(\frac{1}{2}p)) e^{-\frac{1}{2}\tau^2} \\ (T^2)^{\frac{1}{2}p-1} (1+T^2)^{-\frac{1}{2}N} {}_1F_1(\frac{1}{2}N; \frac{1}{2}p; \frac{1}{2}\tau^2 T^2 (1+T^2)^{-1}), T^2 > 0$$

The Bayes estimate for τ^2 is given by

$$(1.14) \quad E(\tau^2 | T^2) = p + (N-1)T^2$$

while the variance of τ^2 given T^2 is given by (de Waal (1974))

$$(1.15) \quad \text{Var}(\tau^2 | T^2) = 2p + 4(N-1)T^2 + 2p(N-1)T^4.$$

2. Bayesian tests of hypotheses

We shall first consider in general the concept of testing several types of hypotheses from the Bayesian point of view (de Groot (1970)). The parameter of concern is the symmetric matrix W .

(a) Simple hypotheses versus simple alternative

Suppose we want to test

$$H_1 : W = w_1$$

against

$$H_2 : W = w_2$$

where w_1 and w_2 are known matrices. The appropriate loss function for this decision problem is

$$L_1(w_1) = L_2(w_2) = 0$$

$$L_1(w_2) = a_1, L_2(w_1) = a_2$$

where $L_i(w)$ means we take decision d_i , $i=1,2$ if the parameter W is w . Decision d_1 implies accept H_1 while d_2 implies accept H_2 . Let $P(W=w_1) = \xi$ be a prior probability with posterior probability given $W = \hat{w}$ as a sufficient estimator for W

$$P(W = w_1 | \hat{w}) = \xi(\hat{w}).$$

The risk if d_1 is taken will be $a_2(1-\xi(\hat{w}))$ and if d_2 is taken $a_1\xi(\hat{w})$.

The Bayes decision will be:

$$(2.1) \delta^*(\hat{w}) = \begin{cases} d_1 & \text{if } a_2(1-\xi(\hat{w})) < a_1 \xi(\hat{w}) \\ d_2 & \text{if } a_1 \xi(\hat{w}) < a_2 (1-\xi(\hat{w})) \end{cases}$$

If we let $f(\hat{w}|w)$ be the density function of $\hat{W}$ given $W = w$, then

$$\xi(\hat{w}) = \xi f(\hat{w}|w_1) / (\xi f(\hat{w}|w_1) + (1-\xi)f(\hat{w}|w_2)).$$

Hence d_1 is taken if

$$\frac{a_2(1-\xi)}{a_1 \xi} < \frac{f(\hat{w}|w_1)}{f(\hat{w}|w_2)}.$$

(b) *Simply hypothesis versus composite alternative*

Suppose

$$H_1 : W = w_0$$

$$H_2 : W \neq w_0$$

where w_0 is known.

A typical loss function for this decision problem is

$$L_1(w) = A(w-w_0)^2 \quad \text{for} \quad -\infty < w < \infty$$

$$L_2(w) = 0 \quad \text{for} \quad w \neq w_0$$

$$L_2(w_0) = B$$

where A and B are positive definite matrices of the same order as W .

Let

$$P(W = w_0) = \xi$$

then the Bayes decision is

$$(2.2) \quad \delta^*(\hat{w}) = \begin{cases} d_1 & \text{if } (1-\xi)AE((W-w_0)^2|\hat{w}) < \xi B \\ d_2 & \text{if } \xi B < (1-\xi)AE((W-w_0)^2|\hat{w}) \end{cases}$$

A matrix $G < H$ iff $H - G$ is positive definite.

(c) *Composite hypothesis against composite alternative*

(i) Suppose

$$H_1 : W \leq w_0$$

$$H_2 : W > w_0$$

where w_0 is known.

We shall assume that the losses $L_i(w)$ resulting from the two decisions d_1 and d_2 are

$$L_1(w) = \begin{cases} 0 & \text{for } w \leq w_0 \\ w - w_0 & \text{for } w > w_0 \end{cases}$$

$$L_2(w) = \begin{cases} w_0 - w & \text{for } w \leq w_0 \\ 0 & \text{for } w > w_0 \end{cases}$$

The Bayes decision is given by

$$(2.3) \quad \delta^*(\hat{w}) = \begin{cases} d_1 & \text{if } E(W|\hat{w}) \leq w_0 \\ d_2 & \text{if } E(W|\hat{w}) > w_0 \end{cases}$$

(ii) Let $H_1 : W \in \omega_1$

$$H_2 : W \in \omega_2$$

where ω_1 and ω_2 are given disjoint mutually exclusive and exhaustive subsets of the parameter space Ω .

The losses are given by

$$L_i(w \in \omega_i) = 0, i=1,2$$

$$L_1(w \in \omega_2) = a_2, L_2(w \in \omega_1) = a_1.$$

Let $P(W \in \omega_1 | \hat{w}) = \xi(\hat{w})$, then the Bayes decision is

$$(2.4) \delta^*(\hat{w}) = \begin{cases} d_1 & \text{if } a_2(1-\xi(\hat{w})) < a_1 \cdot \xi(\hat{w}) \\ d_2 & \text{if } a_1\xi(\hat{w}) < a_2(1-\xi(\hat{w})). \end{cases}$$

But

$$\xi(\hat{w}) = \frac{\xi \int_{\omega_1} f(\hat{w}|w) \xi_1(w) d\nu(w)}{\xi \int_{\omega_1} f(\hat{w}|w) \xi_1(w) d\nu(w) + (1-\xi) \int_{\omega_2} f(\hat{w}|w) \xi_2(w) d\nu(w)}$$

if $\xi_i(w)$ is the prior density of W in ω_i , $i=1,2$ and $P(W \in \omega_1) = \xi$.

Hence d_1 is taken if

$$\frac{a_2(1-\xi)}{a_1 \xi} < \frac{\int_{\omega_1} f(\hat{w}|w) \xi_1(w) d\nu(w)}{\int_{\omega_2} f(\hat{w}|w) \xi_2(w) d\nu(w)}$$

3. Hypotheses on τ^2

(a) Suppose

$$H_1 : \tau^2 = \tau_1^2$$

$$H_2 : \tau^2 = \tau_2^2$$

where τ_1^2 and τ_2^2 are known. The losses are $L_1(\tau_1^2) = L_2(\tau_2^2) = 0$, $L_1(\tau_2^2) = a_1$,

$L_2(\tau_1^2) = a_2$. Let $P(\tau^2 = \tau_1^2) = \xi$, then decision d_1 (accept H_1) is taken if from (1.6), if

$$(3.1) \frac{a_2(1-\xi)}{a_1 \xi} < \frac{f_2(T^2 | \tau_1^2)}{f_2(T^2 | \tau_2^2)} = \frac{e^{-\frac{1}{2}\tau_1^2} {}_1F_1(\frac{1}{2}N; \frac{1}{2}p; \frac{1}{2}\tau_1^2 T^2 (1+T^2)^{-1})}{e^{-\frac{1}{2}\tau_2^2} {}_1F_1(\frac{1}{2}N; \frac{1}{2}p; \frac{1}{2}\tau_2^2 T^2 (1+T^2)^{-1})}$$

If τ_i^2 , $i=1,2$ is large then ${}_1F_1$ can be approximated by (Erdély (1953) Vol. 1)

$$(3.2) {}_1F_1 \cong \frac{\Gamma(\frac{1}{2}p)}{\Gamma(\frac{1}{2}N)} \exp(\frac{1}{2}\tau_i^2 T^2 (1+T^2)^{-1}) (\frac{1}{2}\tau_i^2 T^2 (1+T^2)^{-1})^{\frac{1}{2}(N-p)}$$

Hence d_1 is taken if

$$(3.3) \frac{\hat{a}_2(1-\xi)}{a_1 \xi} < \left(\frac{\tau_1^2}{\tau_2^2}\right)^{\frac{1}{2}(N-p)} \exp(-\frac{1}{2}(\tau_1^2 - \tau_2^2)(1+T^2)^{-1})$$

for large values of τ_1^2 and τ_2^2 .

$$(b) \quad H_1 : \tau^2 = \tau_0^2$$

$$H_2 : \tau^2 \neq \tau_0^2$$

for a known τ_0^2 . The losses are

$$L_1(\tau^2) = a(\tau^2 - \tau_0^2) \quad \tau^2 > 0$$

$$L_2(\tau^2) = 0 \quad \tau^2 \neq \tau_0^2$$

$$L_2(\tau_0^2) = b$$

According to 2.2 d_1 is taken if

$$(3.4) \quad (1-\xi)a E((\tau^2 - \tau_0^2)|T^2) < \xi b$$

$$\begin{aligned} \text{where } E((\tau^2 - \tau_0^2)|T^2) &= \text{Var}(\tau^2|T^2) + (E\tau^2|T^2 - \tau_0^2)^2 \\ &= 2p + 4T^2 + 2pT^4/(N-1) + (p + T^2 - \tau_0^2)^2, \end{aligned}$$

from (1.14) and (1.15).

$$(c) \quad H_1 : \tau^2 \leq \tau_0^2$$

$$H_2 : \tau^2 > \tau_0^2$$

According to (2.2) and (1.14) d_1 is taken if $p + (N-1)T^2 \leq \tau_0^2$.

4. Hypotheses on $\Omega = M'\Sigma^{-1}M$

(a) Consider

$$H_1 : \Omega = \Omega_1$$

$$H_2 : \Omega = \Omega_2$$

According to 2(a) and (1.1) d_1 is taken if

$$(4.1) \quad \frac{a_2(1-\xi)}{a_1 \xi} < \frac{f_1(F|\Omega_1)}{f_1(F|\Omega_2)} \\ = \frac{\text{etr}(-\frac{1}{2}\Omega_1)_1 F_1(\frac{1}{2}N; \frac{1}{2}p; \frac{1}{2}\Omega_1 F(I+F)^{-1})}{\text{etr}(-\frac{1}{2}\Omega_2)_1 F_1(\frac{1}{2}N; \frac{1}{2}p; \frac{1}{2}\Omega_2 F(I+F)^{-1})}.$$

- (b) Consider
 $H_1 : \Omega = \Omega_0$
 $H_2 : \Omega \neq \Omega_0$

According to 3(b) d_1 is taken if

$$(4.2) \quad (1-\xi)AE((\Omega-\Omega_0)^2|F) < \xi B$$

$$\text{i.e. } (1-\xi)AE((\Omega-E\Omega)^2|F) + (1-\xi)A(pI_q + (N-q)F - \Omega_0)^2 < \xi B$$

where $E((\Omega-E\Omega)^2|F)$ is approximately given by (1.5).

- (c) Consider
 $H_1 : \Omega \leq \Omega_0$
 $H_2 : \Omega > \Omega_0$

According to 2(c)(i) and (1.4) d_1 is taken if

$$E(\Omega|F) = pI_q + (N-q)F \leq \Omega_0,$$

- (d) Consider
 $H_1 : \text{tr}_j \Omega \leq t_0$
 $H_2 : \text{tr}_j \Omega > t_0$

for a given t_0 , then according to 2(c)(i) and (1.3) d_1 is taken if

$$\sum_{k=0}^j (N-q)^{(k)} (p-k)^{(j-k)} \binom{q-k}{j-k} \text{tr}_k F \leq t_0.$$

Unfortunately it was not possible to consider the same tests on $\text{tr}_j \Omega$ as those on Ω .

REFERENCES

- ANDERSON, T.W. (1958). *An Introduction to Multivariate Statistical Analysis*. Wiley, N.Y.
- DE GROOT, M.H. (1970). *Optimal Statistical Decisions*. McGraw-Hill, New York.
- DE WAAL, D.J. (1972). On the expected values of the elementary symmetric functions of a noncentral Wishart matrix. *Ann. Math. Statist.* **43**, 344-347.
- DE WAAL, D.J. (1974). Bayes estimate of the noncentrality parameter in multivariate analysis. *Comm. Statist.*, **3**(1), 73-79.
- ERDÉLYI, A. (1953). *Higher Transcendental Functions*. Vol. 1. Mc Graw-Hill, New York.
- JAMES, A.T. (1964). Distribution of matrix variates and latent roots derived from normal samples. *Ann. Math. Statist.* **35**, 475-501.
- NEL, D.G. (1972). On the Moments of the Trace of Noncentral Wishart Matrices and Submatrices. *S.A. Statist. J.* **6**, 93-102.
- STEYN, H.S. and ROUX, J.J.J. (1972). Approximations for the Noncentral Wishart Distribution. *S.A. Statist. J.* **6**, 165-173.

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